Grade 10 Mathematics

November Paper 2 Examination Scope

Friday, 20 November 2026, 08:00 – 10:00 (2 hours)

** Refer to the notes of the following sections. Also refer to the pages and exercises of the textbook that are mentioned in the notes. **

Trigonometry

- ✓ Definitions of ratios
- ✓ Special angles without a calculator
- ✓ Solving sides of right triangles
- ✓ Solving angles of right triangles
- ✓ Angles of elevation and depression
- ✓ Solving triangles (combination problems)
- ✓ Calculator work

Trigonometric Function (Graphs)

- ✓ Sketch the graphs of $y = a.\sin x + q$, $y = a.\cos x + q$ and $y = a.\tan x + q$ for a given domain
- ✓ Find the equation of a given graph (the same functions as above)
- ✓ Apply transformations
- ✓ Interpret transformations
- ✓ Graph interpretation involving: domain and range; maxima and minima; reading off values; vertical and horizontal distances; using the graph to solve equations and inequalities; understand increasing and decreasing parts of a function;

Analytical Geometry

- ✓ Use and apply the distance, midpoint and gradient formulae.
- ✓ Know how gradient links to parallel and perpendicular lines in the Cartesian plane.
- ✓ Collinear points
- ✓ Using analytical Geometry concepts to find equations of straight lines
- ✓ Application of Euclidean geometry theorems to triangles and quadrilaterals in the Cartesian plane.

(Euclidean) Geometry

- ✓ All grade 9 geometry theorems and converses: point, line, triangle (including Pythagoras) and quadrilateral (polygon) geometry
- ✓ Proving triangles similar
- ✓ Proving triangles congruent
- ✓ Parallelogram theorems and converses
- ✓ Midpoint theorem and converse
- ✓ Understand circle terminology: centre, radius and diameter. There will be numerical problems on circle geometry
- ✓ There will be numerical problems, variable problems and proofs on quadrilaterals (e.g. prove that ABCD is a parm)

Measurement

- ✓ Perimeter and area formulae (know, apply and adjust) of basic shapes and composite shapes (made from basic shapes): rectangle, triangle and circle
- ✓ Surface area and volume of formulae (know, apply and adjust) of prisms, cylinder, pyramids
- ✓ Apply and adjust given formulae for SA and V of spheres and hemispheres
- ✓ Use, apply and/or adjust any given formula for measurement.
- ✓ Know and apply conversions within the metric system of measurement
- ✓ Scale factor

Statistics (Data Handling)

- ✓ Measures of central tendency – for grouped (estimated) and ungrouped data: mean, median and mode
- ✓ Class midpoints (for grouped data)
- ✓ Measures of dispersion – for grouped and ungrouped data: range, quartiles and IQR
- ✓ Box and whisker graphs – must be able to work with two graphs.
- ✓ Representation of data (graphs): all types of bar charts, histograms, pie charts, line charts

